

Augmented Reality in Warehouse Operations: Opportunities and Barriers

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Abstract: In today's business environment, the efficiency of warehouses can be critical for the efficiency of the overall supply chains they belong to. As a result, new technologies are being tested and adopted in industry to improve the performance of warehouse operations. An example technology that has recently gained interest by both academia and industry is augmented reality. In this paper, we investigate the opportunities arising from the usage of augmented reality in warehouses as well as the barriers for its industrial adoption. This is done via a series of practitioners interviews and via an experiment designed using Google Glass. Our results indicate that even though the technology is not mature enough at the moment, the potential benefits it can offer make it promising for the near future.

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1. INTRODUCTION

The role of warehousing in modern logistics becomes more and more significant due to several factors such as the rapid increase of e-commerce transactions and the need for reduced inventory and faster response time (Lu et al., 2016). As a result, the effective management of warehouse operations can be critical for the overall performance of the supply chain they belong to. In response to this, several decision support models, that assist warehouse managers in scheduling and controlling their warehouses, have been developed for warehouse operations (Gu et al., 2007).

In the last two decades and with the emergence of computer systems, the deployment of information and communication technologies has often been suggested as an effective solution to managing warehousing operations (Davarzani and Norrman, 2015). Recently, a technology that has caught the researchers' and practitioners' attention with regards to its potential to support manufacturing and logistics processes is augmented reality, an emerging new area for mobile cyber-physical applications (White et al., 2010). Augmented reality involves the combination of the physical and the digital world in real time through a wearable device (Glockner et al., 2014; Ong and Nee, 2013; Cirulis and Ginters, 2013) and has been recognised as an enabler for industrial cyber-physical systems (Leitao et al., 2016; Khalid et al., 2014; Gorecky et al., 2012).

This research study was designed to investigate the opportunities, benefits, barriers and limitations that augmented reality could bring in warehouse management. For this reason, two activities were conducted. The first involved

in depth interviews with practitioners aiming at gaining insight in terms of the current situation and the future of the technology. The second involved a set of experiments with an augmented reality application developed by the authors, aimed at collecting user feedback and examining the issue based on a hands-on experience. The experiments were designed for the sorting process used in warehousing and ran both in a lab setting and in a warehouse trial.

The remainder of this paper is organised as follows. In Section 2, we review existing applications of augmented reality in logistics and warehousing. Section 3 then describes our methodology and data gathering activities in more detail. Section 4 provide some results while Section 5 present the key findings of our study. Finally, we conclude in Section 6.

2. BACKGROUND

In this section we provide a brief review of augmented reality in logistics focusing on its applications in warehousing.

2.1 Augmented Reality in Logistics

Augmented Reality (AR) is a wide term to define a method to add virtual elements, items or information in real time in the physical world. It can be supported by different technologies (e.g. computers, TV, smartphones and tablets, glasses, wearables). The term is not constrained to the visual aspect, since it can also include audio or involve other senses of the user. For this paper, we will focus only on the visual aspect. The definition of an augmented reality system can therefore vary, however, three key characteristics that a system should have can be concluded

from literature (Azuma, 1997; Van Krevelen and Poelman, 2010): The system should *i*) operate in real time, *ii*) mix virtual elements with the reality, *iii*) be integrated in a 3D environment.

In logistics in particular, even though the available literature is quite limited, augmented reality is seen as one of the technologies that could bring the “next big wave of change” in the industry (Glockner et al., 2014). It has also been argued that it can improve the execution of several logistics processes (Cirulis and Ginters, 2013). The interested reader is referred to (Glockner et al., 2014) for a practice-oriented review of use cases in logistics covering warehousing operations, transportation optimization, last-mile delivery, and enhanced value-added services.

2.2 Applications of Augmented Reality in Warehousing

We present our review of existing work in terms of the four main warehouse operations (Davarzani and Norrman, 2015): *i*) receiving, *ii*) storing, *iii*) order picking, *iv*) shipping.

Order picking is by far the most studied area (see for example (ISW, 2012)), perhaps due to the fact that it accounts for more than 50% of warehousing costs (Giannikas et al., 2016). Existing research has focused on how routing of a human operator could be enhanced using AR (Reif and Günthner, 2009; Schwerdtfeger et al., 2009), what is the most effective way to indicate a storage location to a picker (Schwerdtfeger and Klinker, 2008) and comparisons between pick lists communicated by voice, via a head-mounted display, using lights, or on printed paper (Tumler et al., 2008; Weaver et al., 2010; Guo et al., 2014). From an industrial perspective, companies like Knapp,¹ Picavi,² Itelligence and SAP,³ DHL⁴ and Generix⁵ have started developing solutions focusing on different hardware and software elements of an AR solution. The goal is to enable picking which is fast, error free and user friendly while guiding a human operator. Nevertheless, some aspects are missing from industrial solutions, especially concerning barcode readers and real-time 3D projections. Most of the existing solutions with wearable glasses manage only to display the equivalent of printed pick list in front of the eyes of the user.

Relevant studies and solutions for the other three key operations, —i.e. receiving storing, shipping— are more limited (Real and Marcelino, 2011) and they mainly aim to present potential use cases for future developments (Glockner et al., 2014). However, the application of augmented reality in receiving, storing and shipping could potentially lead to benefits similar those observed in order picking, i.e. decreased error rates and faster execution of operations. For this reason and in order to stimulate future areas of work, in Table 1 we present a list of potential uses of augmented reality across the four key warehouse operations.

Table 1. Potential uses of augmented reality in warehouse operations

Operation	Potential uses
Receiving	Indicate the unloading dock to incoming truck driver Check received goods against delivery note Show where to put the items/how to arrange them in the waiting zone
Storing	Inform an operator about a new allocated task Display the storage location of incoming items Display picture and details of the item to be stored Indicate route to storage location Indicate picker's current status as well as next step of the process Check locations requiring replenishment while storing
Picking	Inform an operator about a new task allocated to him Display picture and details of the item to be picked Display the storage location of the item to be picked Display picking route Highlight the physical location with the item required Inform about errors and disruptions Scan the item's barcode to assign to picking cart or to see more information Highlight where to put each item on the picking cart for sorting while picking Give information to prevent congestion in aisles Monitor picker's condition and performance
Shipping	Show what type of cardboard to use Show the best way to place picked items in a package Indicate the right location/pallet for the shipment Show where to place each order on a pallet/in a truck according to type of orders, destination, fragility Indicate appropriate loading area Check/Count products/orders to be loaded on a truck

2.3 Overview of existing work

The conclusions we can draw from our analysis of existing work can be summarised in three key points:

- *Head mounted displays and smart glasses* are of particular interest from an industrial point of view as they offer a hands-free solution. At the same time, they can give the opportunity to the user to communicate effectively with the physical world while wearing them which is crucial in industrial applications.
- Even though, there are a number of proof of concept and conceptual demonstrations of future applications, *a study of the benefits and limitations* of this technology is key before implementation.
- *Order picking* has been the main area of focus although existing solutions focus on routing and picking only, without considering other value-added steps like packing, labelling and sorting.

3. METHODOLOGY

Motivated by the need to examine and understand the potential benefits and challenges from the adoption of augmented reality in warehouses before the implementation of a real industrial system, two activities were designed. The first, a set of interviews with practitioners from the industry was used in order to gain the general insight

¹ <http://www.youtube.com/watch?v=I34Gp0oJ0vI>

² <http://www.youtube.com/watch?v=B6zPnVGS0VI>

³ <http://www.youtube.com/watch?v=0rYHJaSAxis>

⁴ <http://www.youtube.com/watch?v=I8vYrAUb0BQ>

⁵ http://www.generixgroup.com/wp-content/uploads/2015/10/1206814-04-04_Recompense_Prix-Innovatio.pdf

on the issue. The second, an experimental study using an AR app was designed in order to examine the user's perspective in using AR in warehouse operations in a more realistic setting. We describe each activity in more detail.

3.1 Methodology for Practitioner Interviews

A survey through semi-structured phone interviews was conducted with solution providers as well as warehouse managers and experts in the field of logistics and augmented reality. These interviews identified ways augmented reality could be implemented and used in warehouses, discussed its potential benefits, and raised the main concerns expressed in the industry about the technology. Twelve phone interviews were conducted in total with five warehouse managers, five solution providers and two logistics experts from six countries in total (Austria, China, France, Germany, Netherlands and United Kingdom).

The interviewees were asked to comment on both existing solutions and solutions under development. They were also asked to identify those technological features that are/will be required for the adoption and usage of augmented reality in warehousing and how well existing solutions support them. Finally, they were asked to express their views on the future of the technology and its general usability in the future.

3.2 Methodology for Experimental Study

In order to collect more detailed user feedback, a set of experiments were designed that required users to sort packages using a head-mounted display.

Specifically, the warehouse process chosen was the sorting process for packaged orders; this allowed a warehouse company to interact with physical items (packages) they have tagged/label with a label of their choice (e.g. QR code) after packing is completed. At the same time, it provided an extension beyond the areas existing applications have focused on (which normally end with the picking of items). In this study, the user is required to sort several packaged orders in two different sorting bins depending on the last-mile logistics provider that will transport the order to the end-customer. Each bin was indicated by a number (1 or 2) and a marker.

The wearable technology chosen for the experiment was the Google Glass, as a technology a warehouse company would purchase and adopt without the support of third-party companies. Another reason the Google Glass was selected was that it offered a light-weight solution that allowed a user to easily interact with the physical world of the warehouse. For our experiments, two different apps were developed and tested. The first one ("Display App") provides the same information with existing applications that do not use augmented reality: it shows the reference of the sorting bin on a screen by displaying a small sentence and the number corresponding to the bin (See Figure 1). This step is designed so that the user can experience the benefit of the hardware itself without any impact from the software. The second app ("Cube App") adds an augmented reality element: a virtual box appears on the marker linked to the bin while the user is looking at it (See

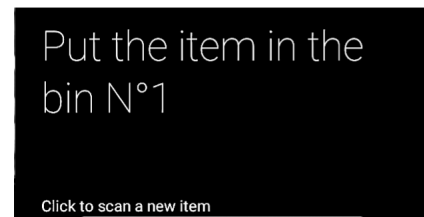


Fig. 1. App view: Display app

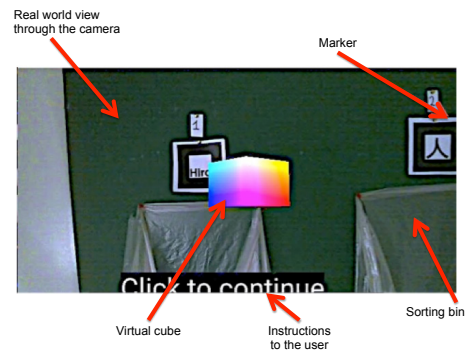


Fig. 2. App view: Cube app

Figure 2). In this second version, the operator is required to look at the right marker before proceeding to the next sorting step and a feedback is given if no bin has been scanned by the Glass. It can be considered as a checking step to automatically verify the process.

The apps were coded with Android Studio in Java language. Two open source libraries were used: ZBarScanner⁶ for the Barcode reader and the ARToolkit⁷ for the AR functionality. The final app is available on Github.⁸

The experiments were conducted in two phases: Firstly, in a laboratory environment with 19 operators. Secondly in a real warehouse with five operators who are familiar with existing non-AR solutions (see Figures 3 and 4). In each set of experiments, each participant was required to sort 10 packaged orders in three trials:

- (1) Using a scanning app on a smart phone that simulates the current technology (a scanner). The scanner was assumed to be attached on a fixed computer.
- (2) Using the "Display App" on the Google Glass. This was designed so that the user can understand the benefits of the wearable technology itself without impact from the software
- (3) Using the "Cube App" on the Google Glass to integrate an augmented reality element in the process by projecting a virtual cube.

Operators were asked to compare the apps using four criteria: preference, ease, speed and error rate impact.

4. RESULTS

4.1 Practitioner Interview Results

During the practitioner interviews, seven features were identified by the interviewees as being very important

⁶ <https://github.com/dm77/ZBarScanner>

⁷ <http://nyatla.jp/nyartoolkit/wp/>

⁸ https://github.com/mhays2/dissertation_stoltz.git



Fig. 3. Sorting bins



Fig. 4. Operator with Google Glass

for the future adoption and success of augmented reality applications. Those features were unprompted responses (i.e. without giving the interviewees a list of features to comment on their importance). Table 2 summarises these and indicates how frequently each one was mentioned.

Table 2. Required features from practitioners

Feature	Description	No.
User interface	Does not require special knowledge to use, user friendly and easy to use, with minimum interactions required	8
Ergonomics	The device is comfortable to wear (balance, weight etc) and does not obstruct the view of the user	6
Scanning	Should be done quickly and in high accuracy using a high precision auto-focus camera, barcode or QR code reader	5
Screen	Big enough to read information, wide-field view, at a natural field of view	3
Battery	No additional device needed to be worn for the battery to last	3
Robustness	The device can be used in industrial environment, (i.e. support dust, can be dropped without breaking etc.)	2
Programming	Easy to program, using a well-established programming language	2

The great majority of interviewees, generally saw good potential for AR in warehouses, especially due to the fact that there are a big number of possibilities and areas that it could be used in. Moreover, the fact that the technology can offer benefits in processes that cannot be automated is of particular importance to practitioners. Even though the technology has already been implemented in certain cases, the battery is still a major technical issue.

In terms of R&D, further research is required regarding the barcode and QRcode readers, the latency and quality of the attached camera, the field of view, the miniaturisation of existing powerful processors that do not overheat and the robustness of the device. It is believed that with the topic being considered by big companies like Microsoft, Google or Amazon, the research will speed up in the next 3-5 years. Also, the price of an AR solution needs to be reduced in order to allow early adoption and use. Finally, an important issue is a change in the existing mind-set of both warehouse managers and operators who need to accept the new technology.

4.2 Results from Experimental Study

The following results per criterion were reported:

- **Preference:** The preference was first for the Cube App, then the Display App and finally the scanner. Nevertheless, some of the reasons were that participants simply enjoyed using new technology. This can be an ephemeral effect.
- **Ease of use:** All participants agreed that the Glass was easier to use than the phone. They mostly agreed on the fact that the Cube App is a convenient way to display the information and it is better than just giving the bin number. It also requires less concentration which, however, can make the overall task tedious. The importance of clear and simple instructions was also noticed.
- **Speed:** The Google Glass apps seemed faster than the phone one. This is mainly due to the fact that by using a Glass, the participants could save time walking between the scanner table and the bins.
- **Error rate:** Participants felt that using the Cube App would reduce the error rate compared to the other two apps as it automatically checks if the right bin is in the view of the operator.

With regards to technological features required for a successful AR solution, participants were asked to identify the three most important ones according to their experience in the experiments. These are summarised in Table 3.

Table 3. Features from experimental study

Group	Top three features
Laboratory operators	A fast and accurate barcode reader A long-life battery A user-friendly interface
Industrial operators	Voice-control or other without need to click Receiving audio instructions besides visual ones A good battery and a fast processor

A number of technical limitations were observed during the experiments. Some of these limitations are device-specific but they can provide recommendations for other devices designed for warehousing applications:

- (1) The distance to read a barcode or QR Code depends on the size of the printed code but also on its quality. As a result, big codes were attached to items.
- (2) The view is not central since the screen is at the top right field of view. This proved to be a difficulty for several users since it is not optimal for the comfort.

- (3) The camera and processor are not designed for intensive use. As a result, the device was running slower, lagging or overheating after a few minutes of usage.
- (4) The Glass is not designed for industrial use, thus making the device not robust enough for warehouses.
- (5) Even in situations when the number of interactions (i.e. clicks) per item was the lowest possible (only one click), it could cause confusion to some users. Voice-control could be examined as an alternative.

Finally, some quantitative results were collected with regards to the speed of each solution. Even though not directly relevant to the objectives of this study, these figures are presented here to demonstrate the relative differences between solutions. Table 4 demonstrates the timing of each solution in terms of *i*) total time required to sort all items, *ii*) time spent to sort each item (waiting excluding) *iii*) time the user took between sorting items. The results indicate that the Display App offer a faster solution compared to the traditional scanner, although the Cube App was much slower. Indeed, the Cube App required more time for the processor to open the camera and show the virtual cube, mainly because it needed a few seconds to wait before doing the next item.

Table 4. Timing per solution (in seconds)

Measure	Phone App	Display App	Cube App
Total	161	137	181
Per item	9	4	10
Between items	6	7	7

5. KEY FINDINGS

In this section, we summarise our findings from the interviews and the experimental study.

5.1 Potential Benefits

The benefits from using augmented reality in warehousing via the usage of wearable technologies like glasses depend significantly on the hardware and results can vary a lot between devices with different specifications. The benefits proposed here do not take into account the technological barriers (which are discussed in the next section). Table 5 presents potential benefits of the technology identified via this study. For each benefit, the table also presents some reasons mentioned by the practitioners, lab testers or operators to justify their opinion.

5.2 Challenges and Limitations

The main barriers identified that could apply to multiple organisations, technologies and business scenarios are summarised in Table 6.

6. CONCLUSIONS

In this study, we investigated the usage of augmented reality in warehouse operations aiming to identify the key potential benefits and barriers for its adoption. Even though augmented reality is not considered to be a new technological development, its usage in manufacturing and

Table 5. Expected benefits from using augmented reality in warehouse operations

Type of benefit	Justification
Reduced error rate	No need to remember the action to complete (portability of information)
	Device shows a picture of the product sought in the field of view
	Limited decision making required (device can show next steps)
	Automatic double checking can be easily done, e.g. with an automated recognition of a product, storage location etc.
	If the operator is disturbed, his next steps are not impacted
More flexibility	Device offers a hands-free solution when an item/package has a big size
	The information can be displayed anywhere in the warehouse and at any time
	If an operator is disturbed, there is no need to walk to a station to check
	With a device proposing central field of view option, it becomes an “eye-free” solution
Improved reliability	Opportunity to share a video or a photo of a defect or issue with a manager not on site
	Less concentration is required as instructions are easily shown to operator
Increased speed	Even if AR solutions are not necessary faster, by decreasing the error rate, re-work is not necessary
	Avoid unnecessary travelling to access fixed computers, carry a scanner etc.
	For certain operations, such as routing, it helps anticipate the moves and result in faster movements (especially for seasonal workers).
Adaptability	The solution does not need a specific environment as the user carries it with him
	It can be suitable for people with disabilities especially with regards to the usage of hands
Safety	Being hands-free, it can be safer for a human operator
	Device can provide feedback and information for safety purposes or warn from an immediate danger
New technology	It brings enthusiasm to many operators especially young ones. This allows acceptance from end user
	From a marketing point of view, it shows that the company adapts itself to latest innovations

logistics operations is still significantly behind compared to other industries like retail and gaming. In this study we identified some of the reasons this is the case which are mainly related to the maturity of the technology for this type of operations and the benefits it can provide compared to existing, well-adopted solutions.

Even though the barriers seem to be significant enough at the moment, with the development of new technologies and the improvement of existing ones, augmented reality might soon be seen in many industrial deployments, perhaps focusing at first at small parts of longer processes. Solution providers and researchers need to keep working closely to the end-user to identify suitable cases augmented reality could be implemented first in order to show real benefits over cheaper alternatives.

Table 6. Barriers for using augmented reality in warehouse operations

Type of barrier	Justification
Hardware limitations	Commercial <i>scanners</i> and smartphone <i>cameras</i> provide a faster and more reliable solution for scanning barcodes and QR codes.
	The <i>battery</i> is not designed to last for long hours in order to cover a full working day. Alternative solutions with extra batteries carried by operators can be cumbersome.
	<i>Processors</i> overheating and slowing down after long periods of use or when complex computing is required can affect the physical process.
	Many wearable, AR devices available are not designed for long period of continuous use which can cause <i>comfort</i> problems: screen latency can cause headaches, spectacles need to be worn by some users, non-central view creates eye tiredness, heavy devices are hard to wear.
Software challenges	Using <i>head-mounted devices</i> , certain operations can be very slow compared to hand-held devices (e.g. checking multiple incoming items).
	<i>Programming environment/languages</i> are not standardised, thus making it hard for practitioners to experiment with devices, develop their own applications and link devices with existing systems.
	With <i>user interfaces</i> being very important for acceptance, simple and intuitive ways to interact with the device are needed to avoid confusion.
	<i>Screens</i> might not automatically adapt to change of light (e.g. from moving in and out a building).
Acceptance	Some users are not willing to wear a device with a camera and mic at all times due to <i>privacy</i> issues.
	<i>Confidentiality</i> issues might arise from the fact that AR devices can capture photos or videos from proprietary operations and data.
Cost	The total <i>cost of ownership</i> is still quite high especially if wearable devices are considered to be personal equipment (i.e. due to hygiene issues) .
	<i>Alternative IT solutions</i> for warehouse management can be significantly cheaper and with well-established benefits.
	Internal IT teams cannot easily <i>maintain and extend</i> AR solutions thus causing extra costs.

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